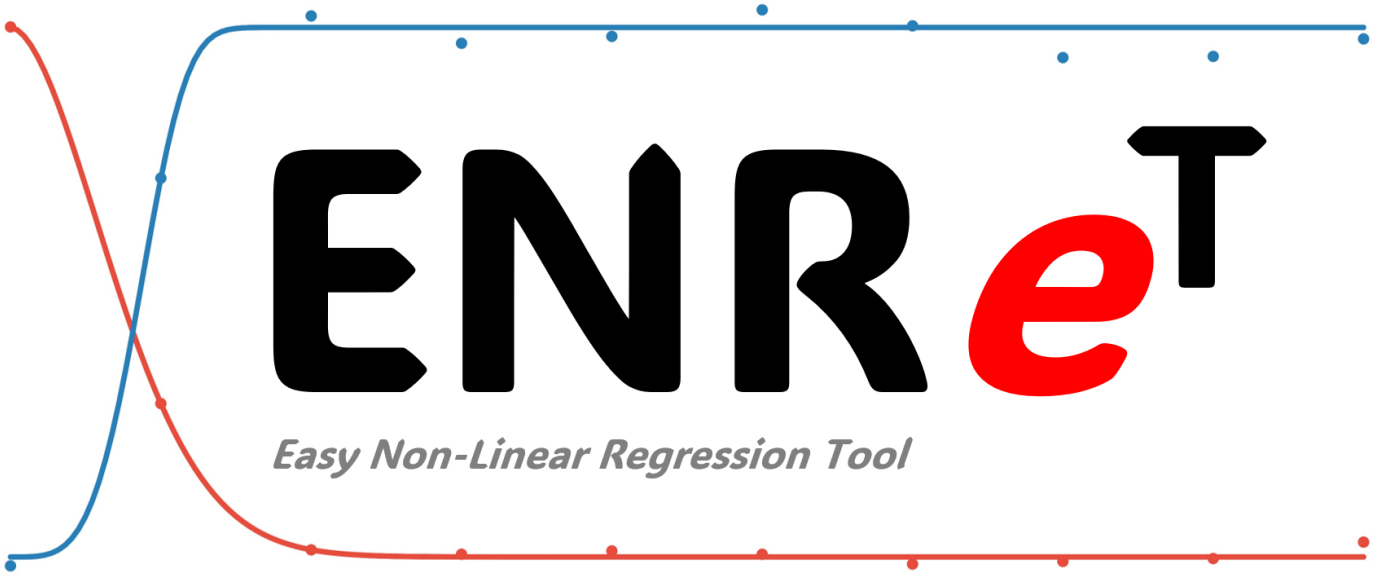


USER GUIDE

A non-linear regression tool for food technologists

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1. Introduction to ENReT

1.1 Document disclaimer

This document was written by me; without AIs. This was a deliberate choice because I wanted this document to feel human, perhaps full of grammatical errors and typos. After working so hard in Vibe Coding, I needed to write something myself, especially to rework and rethink all the work done.

1.2 Background and development with vibe coding

My name is Riccardo De Flaviis, and I have long been interested in modelling, especially applied to Food Science and Technology (FST). The idea of creating a free app/software/tool that could easily help people in this field has always crossed my mind ever since I began self-taught programming. Almost 10 years ago, I created a JavaScript app that could model data from an initial guess using a very simple algorithm called random guesses. The app generated parameters randomly and, following the loss function, oriented itself toward the solution. It was an extremely inefficient algorithm (compared to those available today), but it allowed to find a solution (if the initial guess started very close to the actual solution).

I soon abandoned the idea of developing the app due to the programming difficulty of writing such code. Everything suddenly changed with the arrival of the large language model (LLM) based on the transformer neural network architecture. I have observed and tested AI several times over the years, with mostly negative experiences. I was convinced that AI was not yet ready for my work needs. But collaborating with Dr. Luca Valbonetti (whom I deeply thank) on another project (in 2026) allowed me to observe the coding capabilities that certain AIs had achieved.

With a newfound confidence in these tools, I began a trial, not entirely confident, with Gemini Pro for a nonlinear modelling app. I was surprised to discover that it was possible to create apps even using R, the software I used and still use most often for statistics. This detail was important; writing an app with a programming language I knew was essential to understanding what the AI did and modified, how it built scripts and program logic.

The experience was exceptional, extremely different from previous ones. I recognize that AI has reached an incredible and unexpected level in coding. I gave the prompt of how I wanted the app and the AI created the script. From that moment on, I never stopped implementing the app, always with new ideas and improvements. As the app began to grow, Gemini showed its limits, so much so that I began testing the AI everyone recommended to me, Claude. Claude was a major step forward; the AI was extremely consistent with itself and showed very few hallucinations during development. Of course, there were errors, but they were quickly fixed when reported.

I started out with the idea of challenging the AI for fun, understanding their level, and I ended up, without even realizing it (except after many nice comments from friends and colleagues), developing an app with great potential.

1.3 Mathematical models and Food Science

The field of food science, with all its nuances, must deal with complex systems. Solving these systems is anything but simple. Over the years, food science researchers have developed various mathematical models to predict complex chemical, microbiological, physical, and biochemical phenomena occurring in foods. This intersection of mathematics and food has given rise to a vast number of models; a thriving field, food modelling, especially toward the end of the last century. ENReT fits into this context. Thanks to the coding capabilities of current AIs and the theoretical knowledge developed over the past century, it was possible to build ENReT.

1.4 ENReT: fundamental references and basic features

ENReT (Easy Non-linear Regression Tool) is a free modelling tool developed with the aid of AI, specifically Gemini (Gemini Pro 3.1, Google) and Claude (Sonnet 5 and subsequently Opus 5.5, Anthropic). Although AIs have been widely used to write the code, the theoretical and statistical parts have been based almost exclusively on three fundamental books:

- van Boekel, M. A. J. S. (2008). Kinetic modeling of reactions in foods. CRC press.

- Motulsky, H., & Christopoulos, A. (2004). Fitting models to biological data using linear and nonlinear regression: a practical guide to curve fitting. Oxford University Press.
- Bates, D. M., & Watts, D. G. (1988). Nonlinear regression analysis and its applications. New York: Wiley

Although they are not exactly recent books, they have perfect explanatory clarity, and I recommend them to anyone who wants to delve deeper. There are of course other texts that I consulted, available in the references section, but which had a much lesser impact on the project.

Briefly, my initial goals, later successfully implemented in ENReT, were the following:

1) Perform nonlinear regressions with an intuitive graphical interface (without the need for knowledge of specific programming languages) but without sacrificing the possibilities for advanced graphical and statistical customization.

2) Implement the Monte Carlo approach as an alternative (and more robust) method for calculating parameter errors, with plots that allow even non-experts to understand its advantages.

3) Facilitate successful fitting as much as possible, through a system for visualizing initial guesses, studying the loss function, and algorithms developed for the Batch module.

4) Have the ability to study multiple models or samples simultaneously.

5) Create a list of the most important mathematical functions used in the field of Food Science and Technology in a single app with the option to add custom functions.

6) Create a free tool, oriented towards researchers, professionals, and even students in the field of Food Science and Technology, with a strong focus on optimizing the user experience (UX) and usability.

1.5 Risk, limitations and further developments

ENReT is a tool developed by a single person with the support of AIs. For these reasons, its use must be handled with caution. However, a validation (regarding the fitting) has been performed (the databases used are also available in the app). ENReT has several limitations that could be improved

in future versions (some implementations are not possible if R and/or shinylive are used). The most important are:

1) Possibility to model systems of equations.

2) Possibility to model differential equations.

3) Analyse non-isotherms for models other than kinetic models.

4) Long analysis times for elaboration when computing complicated models/algorithms.

5) Possibility to stop long elaborations without having to reload the entire app.

6) Possibility to model equations by Bayesian approach (MCMC method).

2. ENReT approaches and main structure

2.1 ENReT main suggested approaches

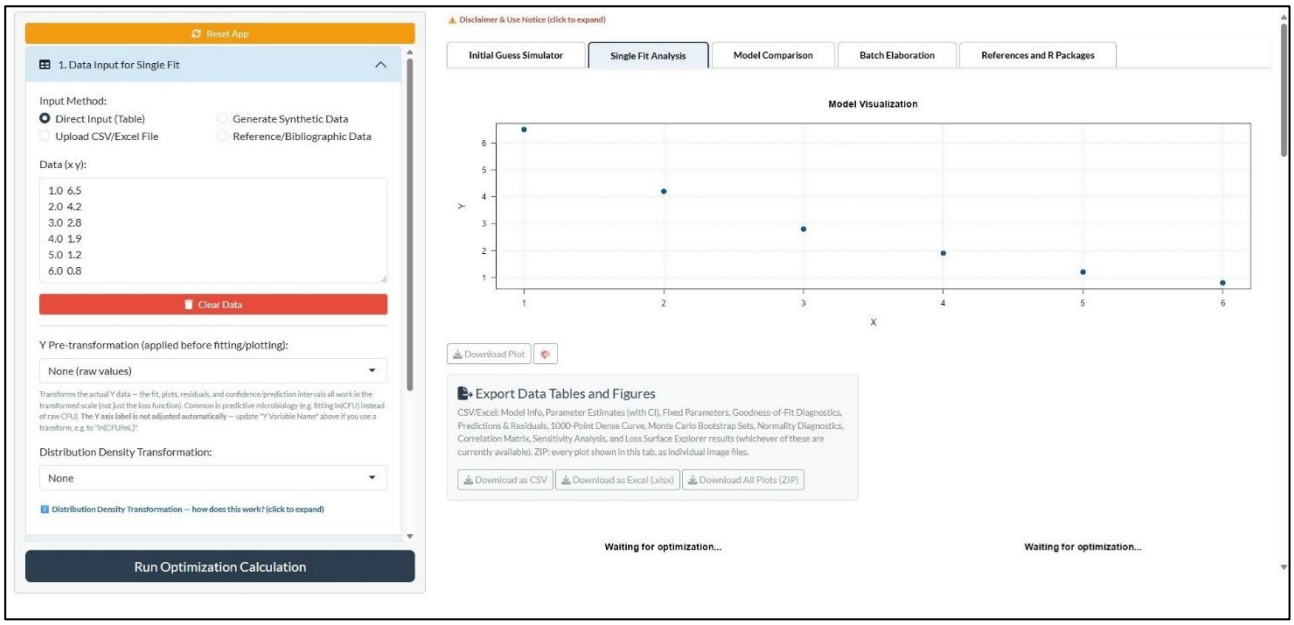
ENReT was designed to perform various tasks for different analysis situations. ENReT can be used in two main ways:

- ***Single-fit approach:*** This approach was developed for those who need to model a few and/or difficult curves. By "difficult," I mean those data points and/or models that cannot be modelled easily and require numerous attempts based on initial guesses or algorithms. The single fit is also recommended for those who want to ***compare different models*** and/or different estimation methods of the parameter errors (three methods available). This type of analysis can be performed via the Single Fit Analysis tab and subsequently with the Model Comparisons tab.
- ***Batch approach:*** This approach is only recommended for those who have many samples to analyse with the ***same model***. This type of analysis can be performed via the Batch Elaboration tab.

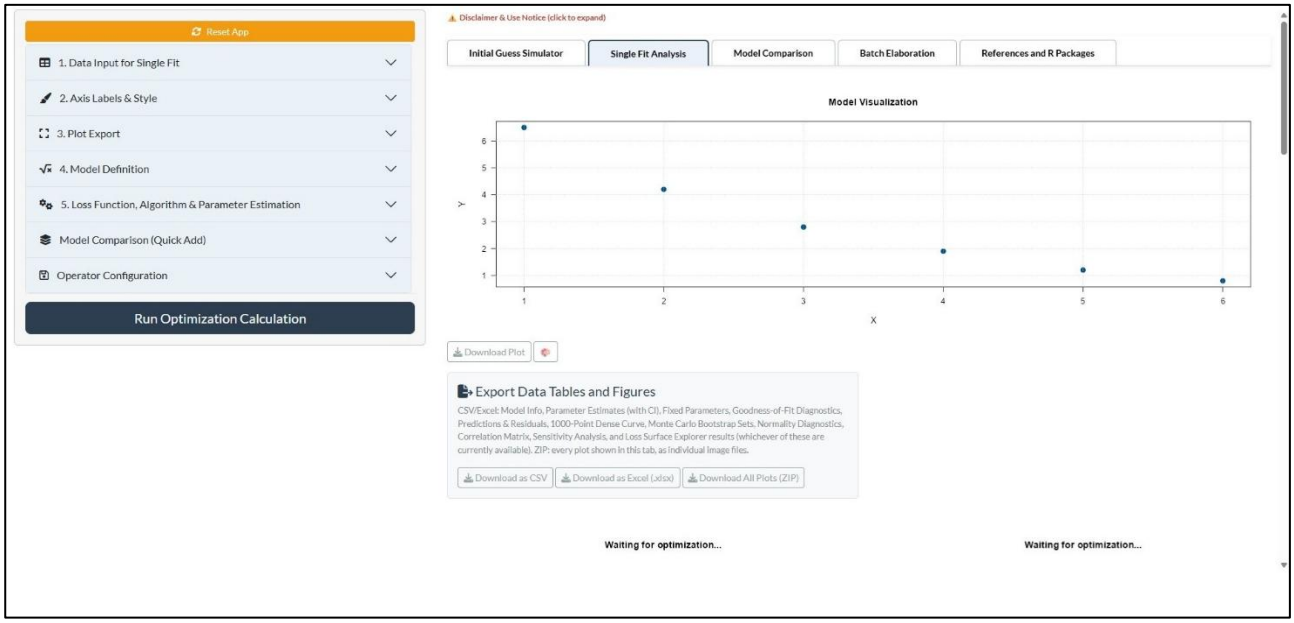
Both approaches have several features in common and differ in some additional implementations (especially in the case of single-fit).

2.2 Main screen and tabs

ENReT is essentially a Graphical User Interface (GUI), and the main screen starts as follows:



The core of the app is 4 tabs selectable at the top, namely “Initial Guess Simulator”, “Single Fit Analysis”, “Model Comparison”, and “Batch Elaboration”. In each tab various additional analyses can be performed, with distinct modules for advanced analyses. The sidebar on the left instead has 7 modules (as shown in the next figure) useful for setting up different analyses and customizations.



2.3 Sidebar modules

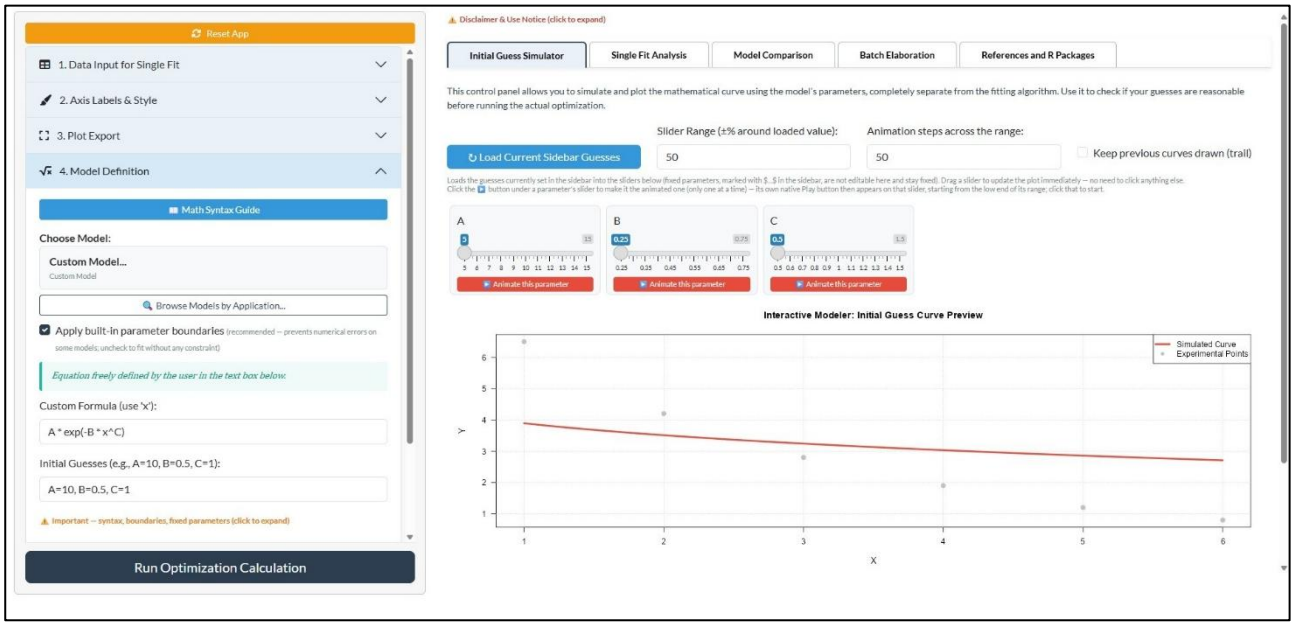
These modules located on the left serve as a central node for setting up various customizations. Below main features of each module are listed:

- ***Data Input for Single Fit:*** this module is only useful if a Single-fit approach (see section 2.1) is performed through the Single Fit Analysis tab (see section 2.5). Here data points can be uploaded in different ways: by simply copying and pasting in two columns format (x and y); or by uploading via CSV/Excel file always organized in two columns (x and y). The app recognizes headers if they are not numbers. ***Warning: if the x or y labels are numbers, the app will treat them as data points and not labels.*** It is possible also to pre-process the data by same basic mathematical manipulations (e.g. Log, Ln, exp) by selecting one option in Y Pre-transformation. Moreover, specifically for particle size distribution performed using sieves, an additional option has been added that allows the data to be transformed into distribution density data. An advanced option available here is the ability to generate synthetic data using the appropriate radio button. With this algorithm, it is possible to create data with a certain noise level and a specific number of replicates based on the model currently selected in the Model Definition module below. Lastly, a series of public data used to validate the app have been added. These datasets can be used by users to test ENReT.
- ***Axis Labels & Style:*** here the aesthetic options for customizing your graphs can be found. Specifically, the x and y names can be changed, also by using Greek letters, superscripts, and subscripts (see the Label Syntax Guide for the specific syntax required). The "Plot Aesthetics & Colors" button allows to access to all the customization options for each graph. There is no need to use this button, as each graph has its own customization button. ***Warning: only here will it be possible to remove the title and background grid from each graph.***
- ***Plot Export:*** export settings of all graphs; available in two formats, vector PDF or JPEG.

- ***Model Definition:*** this module allows to define the mathematical model of interest or define a custom one. The syntax is defined in the Math Syntax Guide. A wide range of models typically used in the field of Food Science & Technology can be selected. For each model an initial guess and some boundaries are set by default in order to help the user. ***Warnings: the default initial guesses and boundaries may be incorrect for the User's specific problem.*** Moreover, it is also possible to fix some parameters to a known value. The syntax for defining boundaries and fixing a parameter is defined in the module, specifically by clicking in "Important - syntax, boundaries, fixed parameters (click to expand)". ***Warnings: boundaries are accepted only by the Levenberg-Marquardt and L-BFGS-B algorithms, which will be defined in the next module. Other algorithms will ignore parameter boundaries.***
- ***Loss Function, Algorithm & Parameter Estimation:*** this module allows for various advanced modelling customizations. The user can choose from several loss functions and optimization algorithms. ***Warnings: modify these parameters only with the appropriate theoretical knowledge; the default parameters can be considered sufficient for most applications.*** Furthermore, in this section, the user can choose whether to perform a Monte Carlo analysis instead of the classic linear approximation. This choice has important repercussions throughout the app and significantly affects processing times.
- ***Model Comparison (Quick Add):*** this module allows to automatically load the optimization result into the Model Comparison tab for further advanced analysis. By default, any modelling done will be sent to the Model Comparison module.
- ***Operator Configuration:*** this module is important for those who plan to use ENReT many times, in fact it will be possible to save all the settings made within the work session and reload them in other sessions.

2.4 Initial Guess Simulator tab

This tab is useful for understanding the mathematical behaviour of certain equations and the impact of each parameter on the fit. It is also an excellent way to determine the correct initial guesses to provide to the algorithm. As a first step, the mathematical model and initial guesses should be loaded from the sidebar using the appropriate button at the top (Load Current Sidebar Guesses). Subsequently, the parameters can be modified as desired, and the curve will be plotted. The curve can be also animated by moving the parameters by clicking on Animate this parameter. By tracing each attempt (by selecting Keep previous curves drawn check box), it will also be possible to better plot the movement of the curve as the parameter varies.

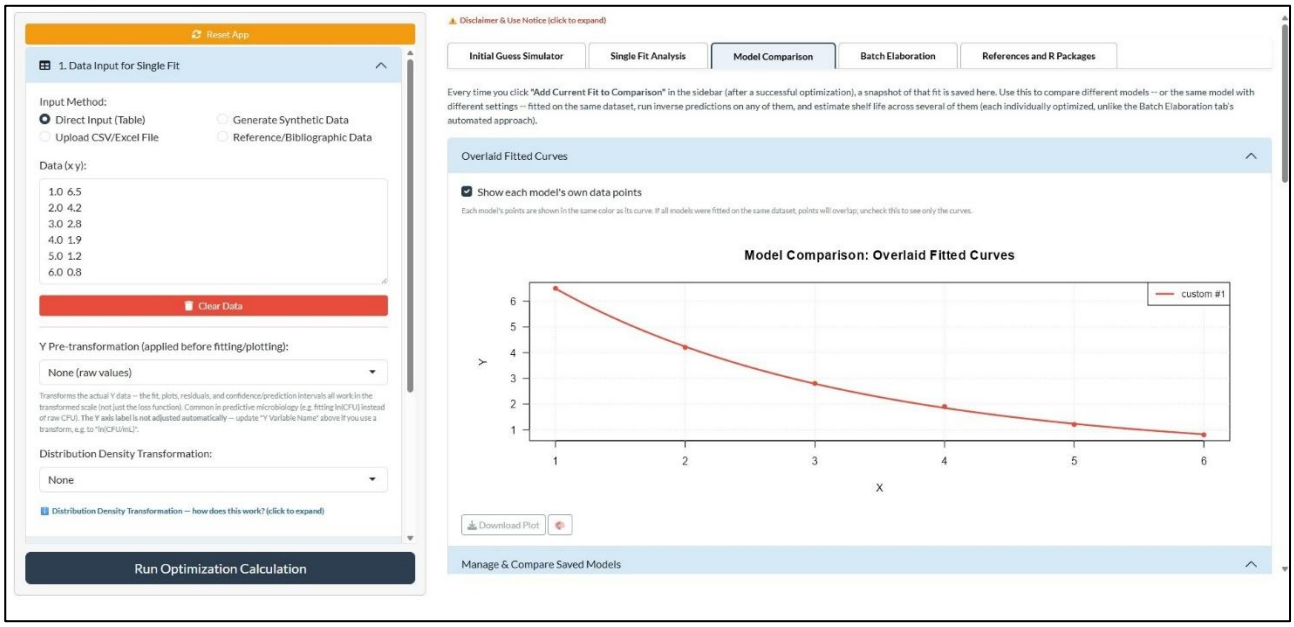


2.5 Single Fit Analysis tab

This tab allows to model each datapoint series individually on the basis of the settings selected in the sidebar. To model data, simply select the button at the bottom left "Run Optimization Calculation". Once the modelling is complete, the app will notify the user whether the fitting was successful or not via a pop-up at the bottom right. The tab automatically generates several diagnostic tables and graphs, briefly: residual plot, Q-Q plot, lag plot, runs test, Parameter Estimation table, Goodness-of-Fit Diagnostics table, Model Predictions and Residual Analysis Table, Parameter Sensitivity plot, and Parameter Correlation Matrix. ***Warning: some plots and diagnostics are available only if a Monte Carlo optimization is carried out.*** Lastly, it is possible to calculate the error of the parameters by a third algorithm, namely Profile likelihood (see Bates & Watts, 1988). To do this, simply click on the "Run profile likelihood" button in the relative module of this tab.

2.6 Model Comparison tab

This tab requires at least two successfully modelling runs in the Single Fit Analysis tab. This tab permits to compare different models fitted with the same dataset or different data modelled with the same model. All completed fittings will be automatically uploaded here (you can also disable this option from the sidebar), however fittings can be deleted or renamed as needed through the Manage Models module available immediately after the first plot (see the figure below).



Subsequently, three tables will be automatically displayed: Comparison Table, Pairwise Evidence Ratio Matrix, and Parameter Estimates Comparison table. Moreover, it will be possible to perform statistical significance tests between the same parameters by clicking on the “Run Test” or “Run Post-Hoc Comparison” buttons. This tab features 3 additional advanced modules, present at the bottom, briefly explained below:

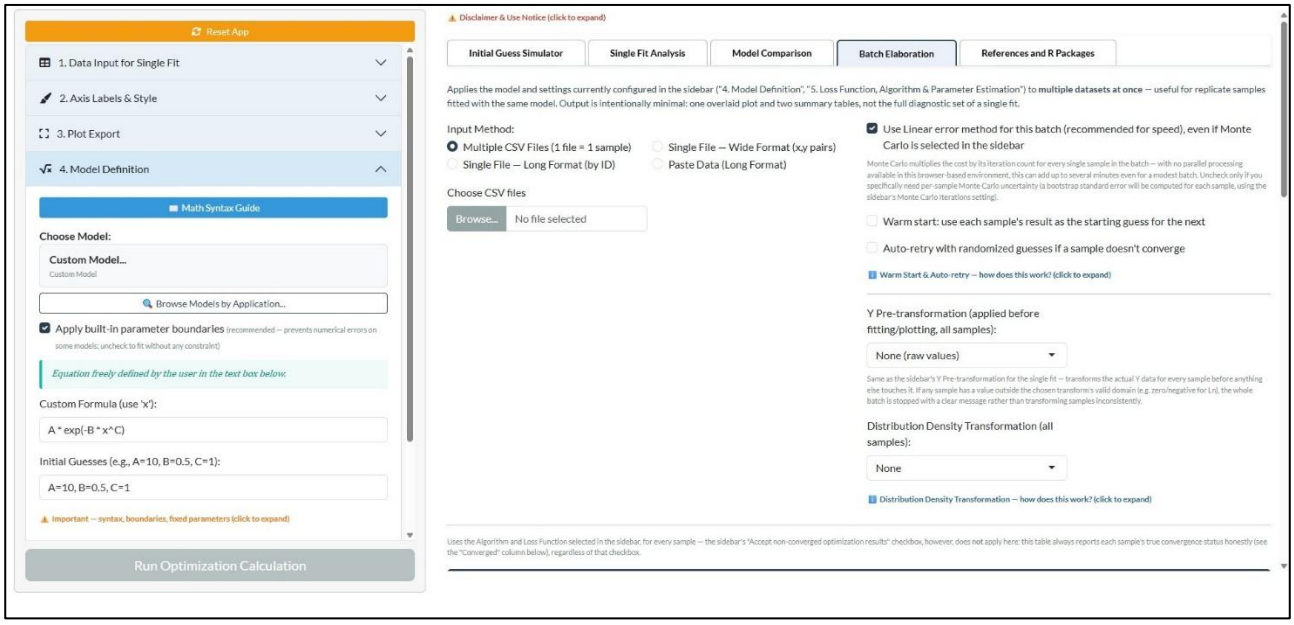
- Inverse Prediction: allows to find x and its error based on a given fixed y (e.g. threshold limit).
- Shelf-Life & Kinetic Model Analysis: allows to determine an x (which in this specific case becomes the shelf-life time) at a temperature different from that of the original tests. It requires at least 3 fittings at 3 different temperatures.

- Non-Isothermal Kinetic Prediction: allows you to predict kinetics with non-uniform temperatures. Currently, this module can only be used with zero-, first-, second-, and n-order kinetic models.

Warning: secondary models (in both shelf-life and non-isothermal modules) are fitted using the Levenberg-Marquardt algorithm and Loss MSE regardless of what has set in the sidebar.

2.7 Batch Elaboration tab

This tab is almost a slimmer copy of the previous 2 tabs, developed to be able to process multiple samples at once. ***Warning: some advanced analysis and plots (e.g. profile likelihood) are available only by using the Single Fit Analysis tab.***



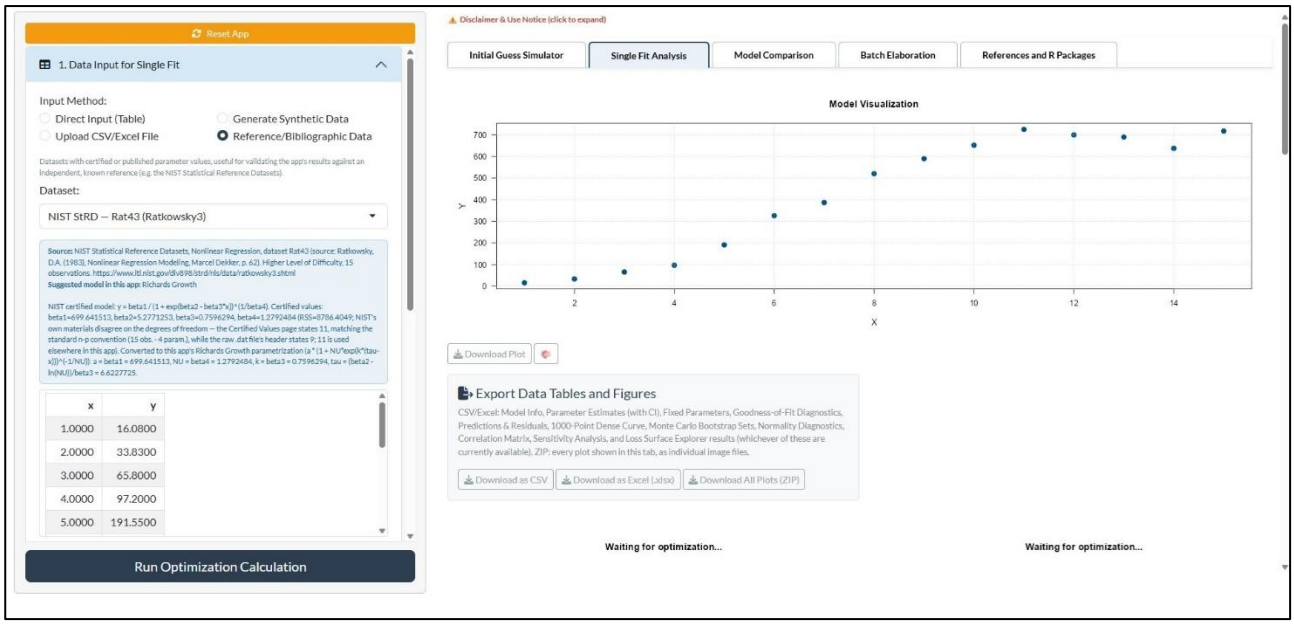
An algorithm (e.g. “Auto-retry with randomized guesses if a sample doesn't converge”) has also been implemented that attempts to modify the initial guess in case the modelling fails. It is a crude algorithm, created without following authoritative sources, and should be used with caution. ***Warning: An additional check was added for the Monte Carlo via the " Use Linear error method for this batch (recommended for speed), even if Monte Carlo is selected in the sidebar" check box. This is necessary to prevent users from accidentally starting a long batch Monte Carlo analysis.***

3. Guided examples

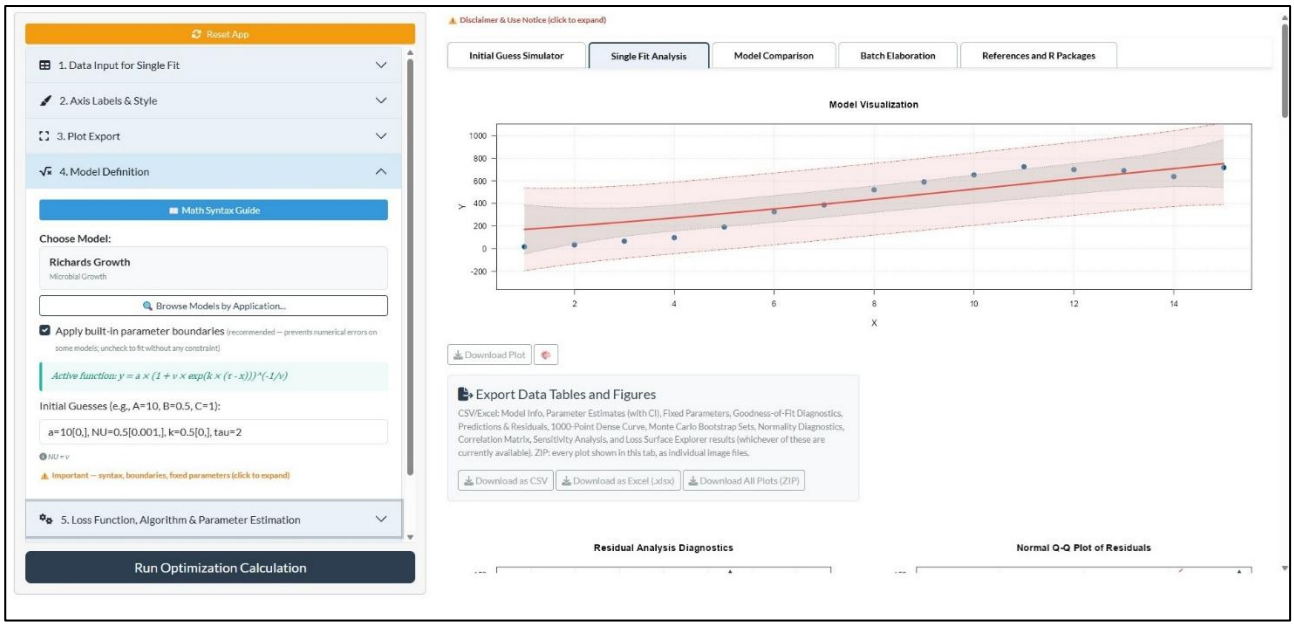
To help anyone approach ENReT, two distinct examples are provided. These examples are only to demonstrate ENReT capabilities, they have no educational purpose.

3.1 Single fit approach

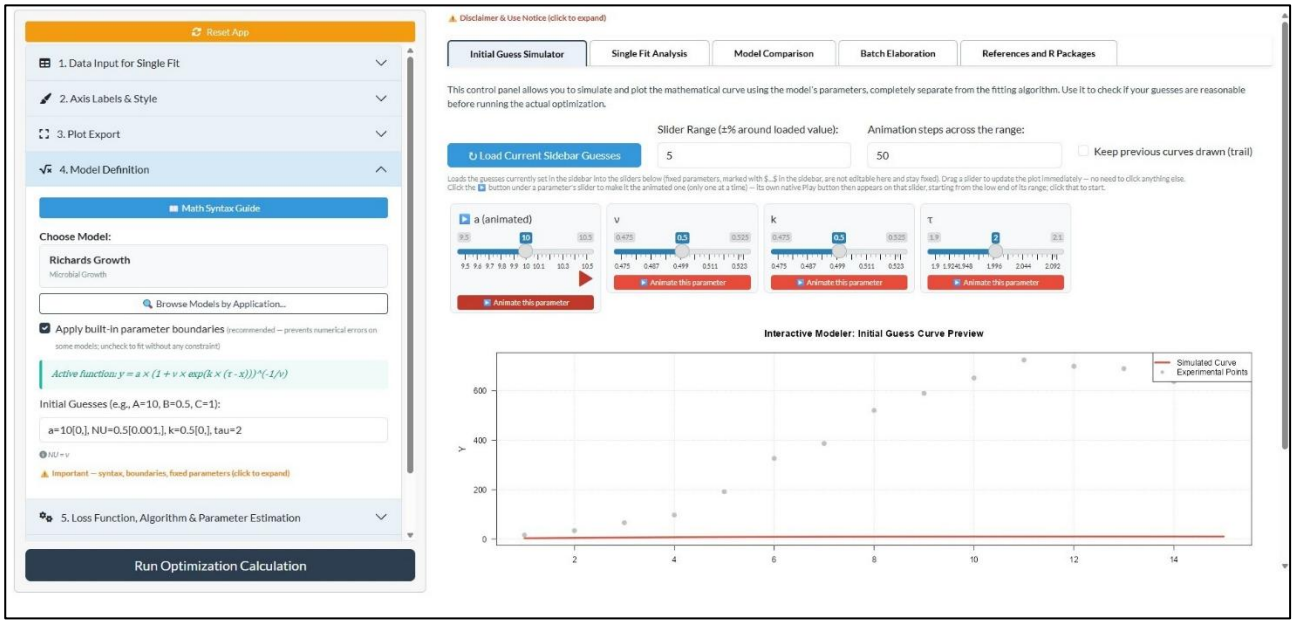
For this example, a public dataset available also in the app is used, namely the NIST StRD - RAT43 to be modelled with the "Richards Growth" model (Microbial Growth). The User can load the dataset from the radio button “Reference/Bibliographic Data” as shown below.



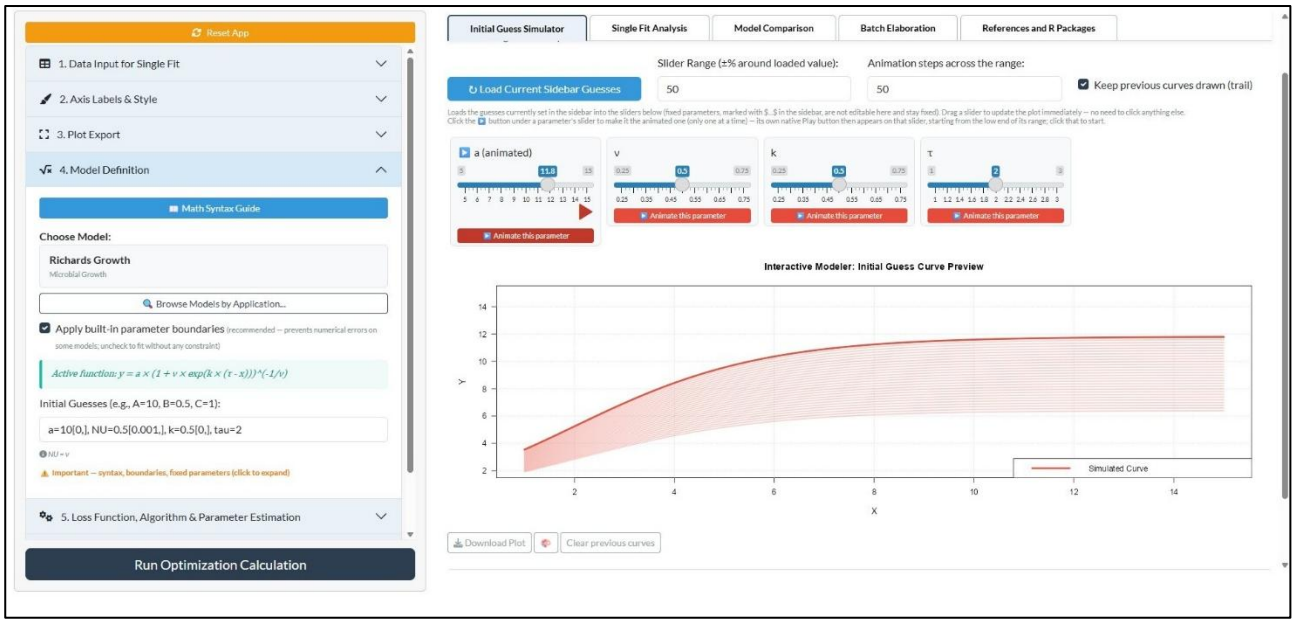
For this example, the " Richards Growth" model will be used (the user could also select the associate model "NIST Rat43" in the category "Models for Reference Data"). To perform a "classic" fitting, the "Run Optimization Calculation" button can be already clicked. The result is not very convincing (see the figure below), and this can be clearly seen from the diagnostic plots (i.e. the residuals and the Q-Q plot). Other warning signs are the high MSE value and standard errors of the parameters. In many cases, a poor fit is due to an incorrect initial guess. Therefore, it is possible to better understand the quality of the current initial guess using the Initial Guess Simulator tab.



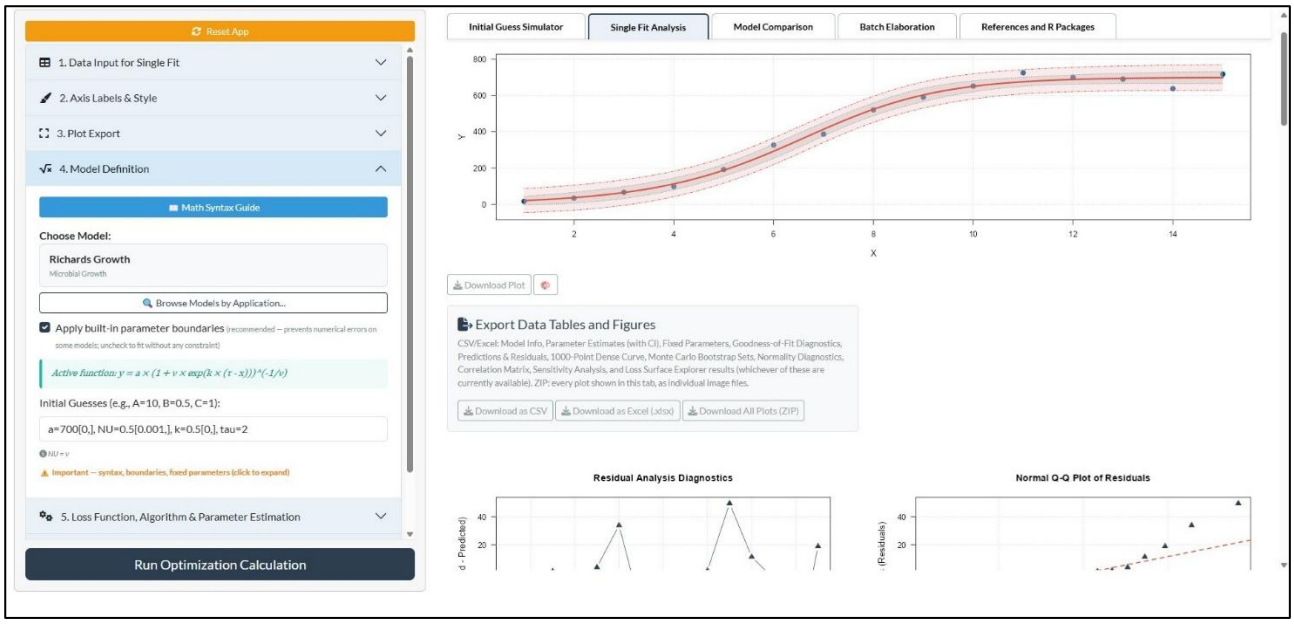
As shown in the next figure, the initial guesses are very far from the raw data.



If the user is unfamiliar with the behaviour of the parameters in this equation, the raw data can be removed from the graph options (by clicking the colour palette button) and then the user can modify the parameters to understand their meaning. By moving the parameter "*a*" it is possible to understand how it affects the maximum of the function as shown the figure below.

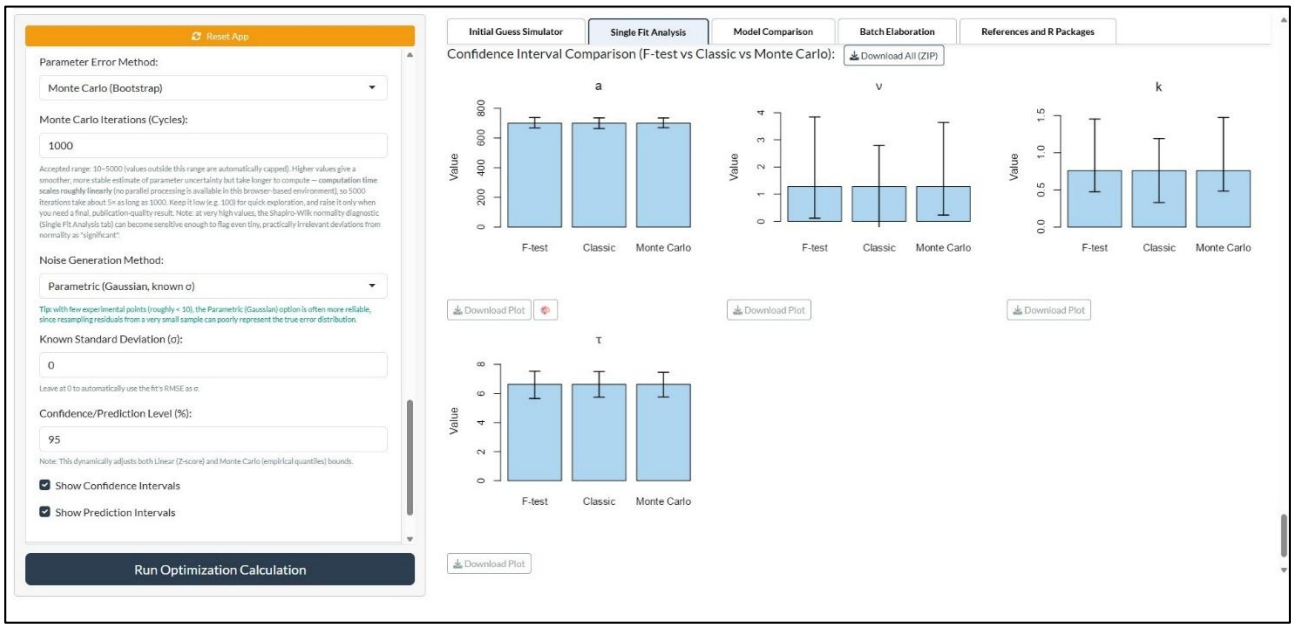


Now it is much clearer that the "*a*" parameter is most likely too far from the optimum (the maximum y of the raw data is 724, almost 2 orders of magnitude more). Now let's go back to the Single Fit Analysis tab and try replacing the initial guess for "*a*" with 700 and restarting the fitting ("Run Optimization Calculation" button). Finally, a good fitting is obtained as shown below.



Looking at the errors in the parameters, one of the (many) limits of the linear approximation in modelling is easily visible, in fact the confidence interval of the ν parameter starts from negative values, which is theoretically impossible on the basis of the boundaries set. A possible solution to this problem is through the Monte Carlo approach. By performing a Monte Carlo analysis (1000 cycles),

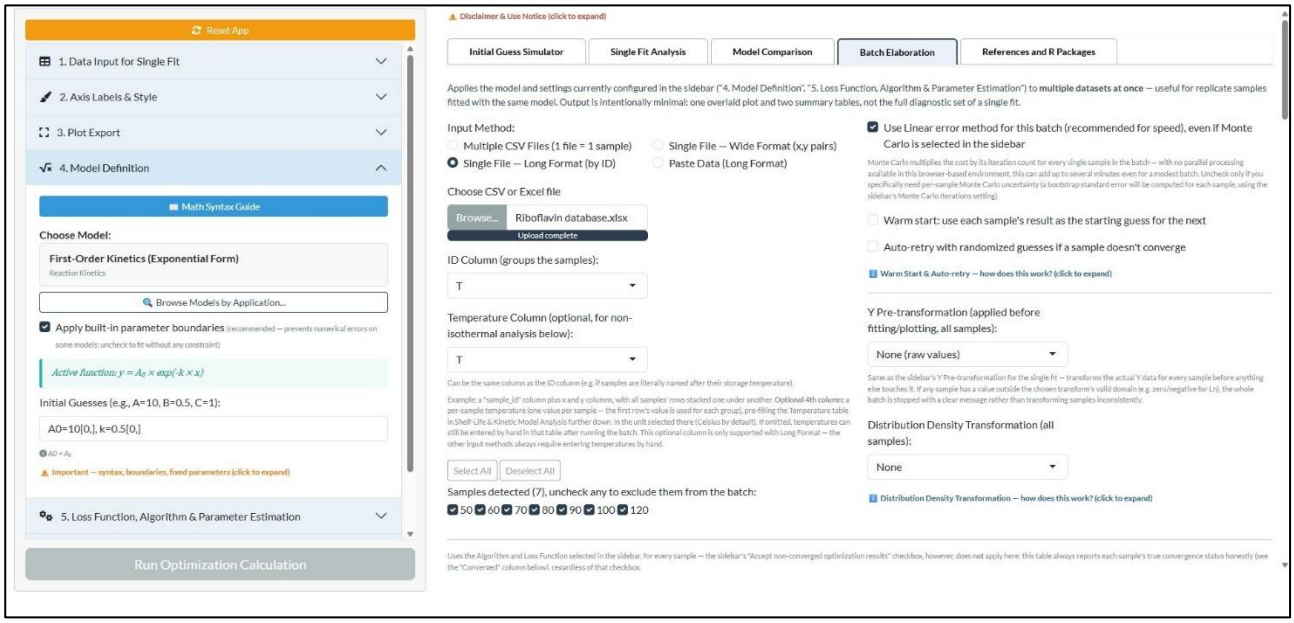
it is possible to obtain a correct confidence interval, as can be seen from the bar plots below where the errors of the three methods available in the app are compared.



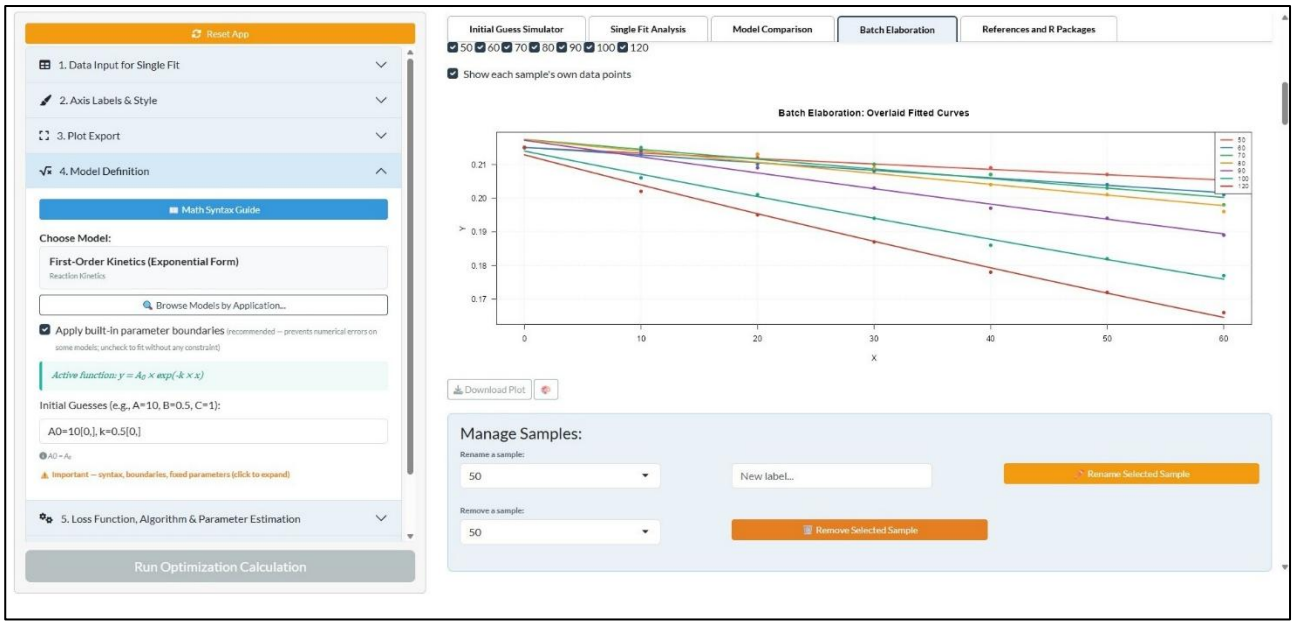
The three methods reveal different information, but it is interesting to see how the profile likelihood and Monte Carlo approach allows to highlight the skewness in the errors of some parameters (i.e. ν and k).

3.2 Batch approach

For this example, a dataset available in Van Boekel (2008) is modelled. The data are published by Nisha et al. (2005) and refers to isothermal heating (7 different temperatures) on riboflavin content in spinach. For this example, the "First-Order Kinetics (Exponential Form)" model will be used. The dataset is organized into 3 columns: x, y, and temperatures (which for ease will also be used as the Sample ID). However, it will be possible to define which column is responsible for the ID samples with the appropriate selector (see the figure below).



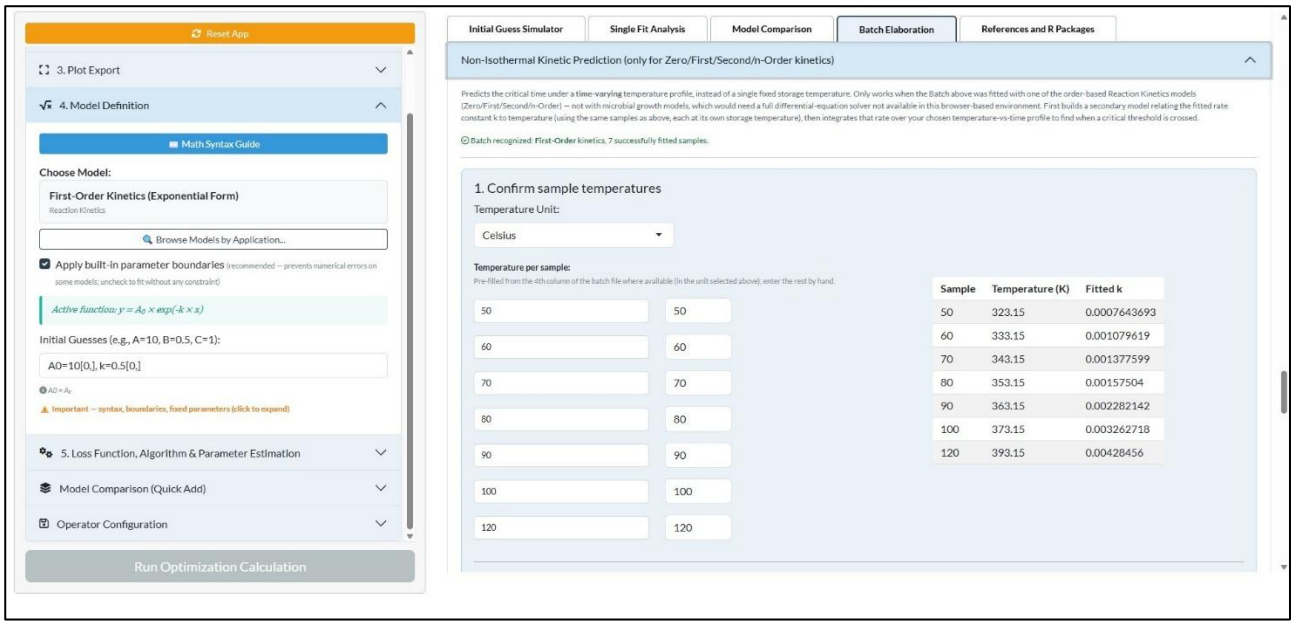
Once the data has been loaded and the primary model chosen by the sidebar, it will be possible to automatically model all the samples (7 in total) using the "Run Batch Elaboration" button. The fittings of all curves are done quickly and successfully as shown in the figure below. Even the MSE values and the errors in the parameters seem to confirm this. Using the Post-Hoc Comparison (3+ Samples) module the parameters of each temperature can be compared with each other by clicking the “Run Post-Hoc Comparison” button.



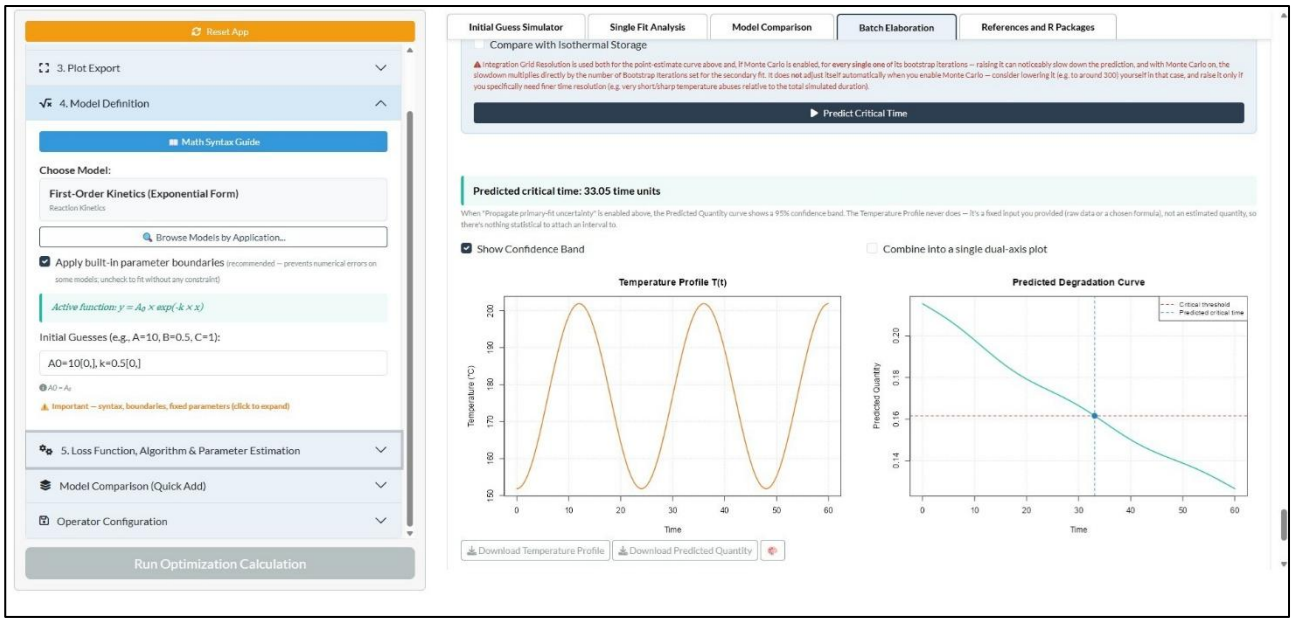
As expected, the A parameters are all statistically non-different while the k parameters show several differences.

Parameter	Sample	Value	Group
A_0	50	0.21499	a
A_0	60	0.21509	a
A_0	70	0.21744	a
A_0	80	0.21736	a
A_0	90	0.21716	a
A_0	100	0.21397	a
A_0	120	0.21284	a
k	50	0.00076	a
k	60	0.00108	b
k	70	0.00138	ab
k	80	0.00158	bc
k	90	0.00228	c
k	100	0.00326	d
k	120	0.00428	e

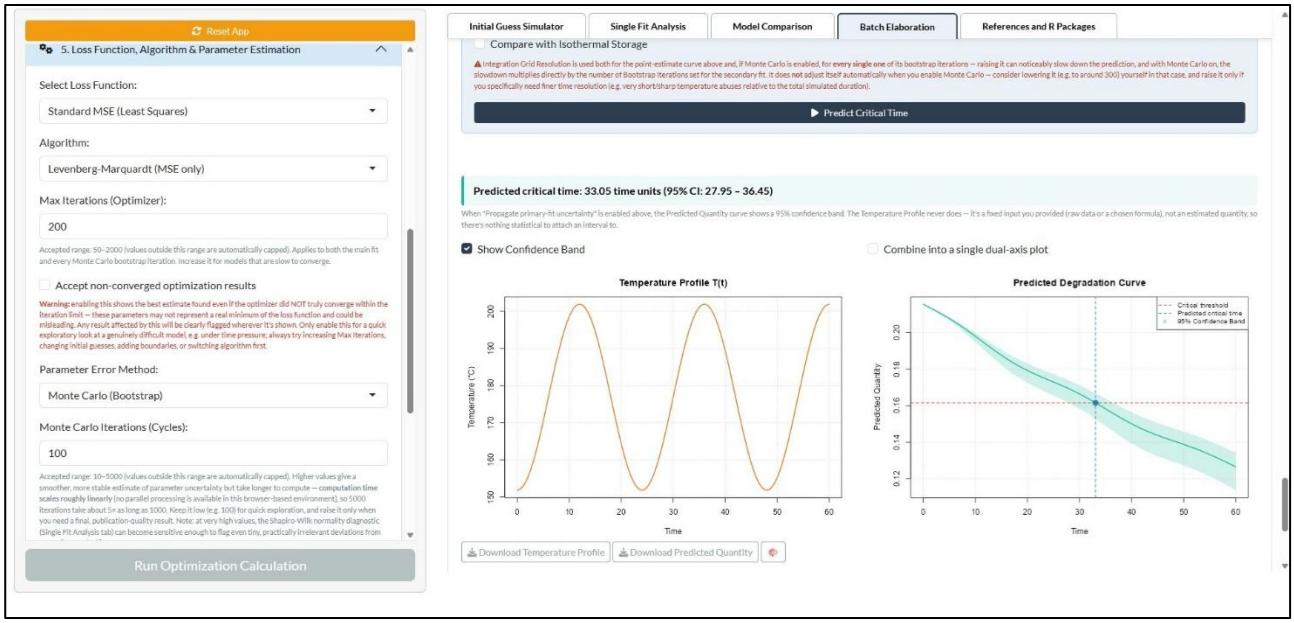
The next step will be to use the non-isothermal module to make a prediction for varying temperatures. If temperature was initially provided, this information will automatically be provided to this module as shown below.



Once the correct temperatures of each sample have been verified, the secondary fitting can be carried out using Log-Logistic (any other model or a custom one can be used) by clicking on the "Fit Secondary Model" button. Once the successful fitting has been verified, it will be possible to carry out the final non-isothermal analysis. For this example, a sinusoidal formula to simulate a hypothetical non-linear temperature trend is used, the parameters are set as follows: Tm=450 (in Kelvin), a=50 and L=24. The rest of the parameters can be left as default. After clicking on the "Predict Critical Time" button it is possible to examine the results as shown in the next figure. With this thermal profile we obtain our limit value (critical Y) in 33.05 minutes of treatment. From the graph on the right, it is also possible to observe the impact of a very strong thermal gradient on the curve (obviously an unrealistic situation created only for demonstration purposes).



It is possible to carry out the entire analysis through the Monte Carlo approach starting from the primary fitting and obtaining the confidence intervals and bands as shown in the last figure.



4. Final remarks

This guide is a concise summary of ENReT. I did not want to create a 1000-page book that no one would read. I deliberately avoided all the mathematical, statistical, and algorithmic sections to streamline the process. Most of this type of information is presented in detail in the bibliography available at the end of this guide. However, I recognize that these details are useful to those who do not have the time (and/or desire) to dig through the app's code. I hope to create a truly comprehensive manual in the near future that also provides the technical details.

On the other hand, lately I have discovered several bugs that have emerged with use and time and that the AI had not noticed. With such a big project, even Claude AI struggles.

That said, I hope to continue implementing ENReT over time, test it in an attempt to find bugs and errors, perhaps also based on your suggestions.

My email: *enret.app@gmail.com*

References

- Barbosa-Cánovas, G. V., Fontana Jr, A. J., Schmidt, S. J., & Labuza, T. P. (2020). Water activity in foods: fundamentals and applications. John Wiley & Sons.
- Bates, D. M., & Watts, D. G. (1988). Nonlinear regression analysis and its applications. New York: Wiley.
- Bourne, M. (2002). Food texture and viscosity: concept and measurement. Elsevier.
- Burnham K.P., & Anderson D.R. (1998). Model Selection and Inference. A Practical Information and Theoretic Approach. New York: Springer Verlag.
- Carroll, R. J., & Ruppert, D. (1988). Transformation and weighting in regression. New York: Chapman and Hall.
- Davison, A. C., & Hinkley, D. V. (1997). Bootstrap methods and their application. Cambridge: Cambridge University Press.
- Efron, B., & Tibshirani, R. J. (1993). An introduction to the bootstrap. New York: Chapman & Hall.
- Gardini, F., & Parente, E. (2013). Manuale di microbiologia predittiva. Italia: Springer-Verlag.
- Kilcast, D., & Subramaniam, P. (2011). Food and beverage stability and shelf life. Elsevier.
- Motulsky, H., & Christopoulos, A. (2004). Fitting models to biological data using linear and nonlinear regression: a practical guide to curve fitting. Oxford University Press.
- Neter, J., Kutner, M. H., Nachtsheim, C. J., & Wasserman, W. (1996). Applied linear statistical models.
- Nisha, P., Singhal, R. S., & Pandit, A. B. (2005). A study on degradation kinetics of riboflavin in spinach (*Spinacea oleracea* L.). *Journal of food engineering*, *67*(4), 407-412.
- Peleg, M. (2006). Advanced quantitative microbiology for foods and biosystems. Contemporary Food Science. CRC Taylor & Francis.
- van Boekel, M. A. J. S. (2008). Kinetic modeling of reactions in foods. CRC press..

R packages

- R (v4.6.1)
- shiny (v1.14.0)
- minpack.lm (v1.2.4)
- bslib (v0.12.0)
- writexl (v2.0.1)
- zip (v3.0.2)
- readxl (v1.5.0)

AI for coding ENReT

- Google. (2026). Gemini Pro 3.1 [Large language model]. https://gemini.google.com
- Anthropic. (2026). Claude (Sonnet 5 and Opus 5.5) [Large language model]. https://claude.ai